R106/A_frozen | Native adaptation and actual latency

Real online native+selector; GPU-resident batch128 amortized, not singleton or I/O end-to-end.

Problem	Instances	Online milliseconds / instance
TSP	128	0.451535
CVRP	128	0.568175
ATSP	128	1.161727
OVRP	128	0.169895
VRPB	128	0.169072
VRPL	128	0.169299
VRPTW	128	0.170011
OVRPTW	128	0.169358
OVRPB	128	0.170226
OVRPL	128	0.169591
VRPBL	128	0.168933
VRPBTW	128	0.169376
VRPLTW	128	0.168772
OVRPBL	128	0.168634
OVRPBTW	128	0.168821
OVRPLTW	128	0.168771
VRPBLTW	128	0.169037
OVRPBLTW	128	0.168831

Total: 11,965,097; trainable: 1,941,257; native trainable: 0

R106 compares frozen A with trainable B native encoders under the same final original selector supervision. Both start from identical complete fresh R84 seed2 selector tensors, three projections and original R102B pretrained ELG/MatNet/ReLD weights. Native pretraining is not selector continuation; its data/overlap remains unknown. Total tensors are identical at initialization, but trainable freedom and training compute are NOT matched. Native stays eval in both arms; B receives gradients from update1 with effective LR0.1 times the common warmup/cosine LR. A does not update native parameters. One input-only native graph per microbatch is shared by the original two dropout selector forwards, with the unchanged CE/comparison/risk/R-Drop and real-row normalization. No auxiliary head or target. All TRAIN/VAL/TEST/LIB calls use online native encoders, never native-feature cache substitution. Native FP32 output precedes normal fusion/selector precision. Raw node4/joint2, physical TSP node0 role, methodID+hand34 and score MLP/maximin remain. Padding-adapted native execution is shared across arms; numerical fixture agreement is not bitwise equivalence or efficacy. Native/FP64 ind, legal-pool Oracle, full-TRAIN SBS; ALL is equal-task macro, mixed raw cost descriptive only. The fixed R84 target36 reports true rescues/losses, false picks and aligned net cost, not choice frequency as recovery. R94B and completed R102B recovery report view are historical, NOT newly matched controls; external pretraining and historical compute are not controlled. Development-exposed TEST; one whole18 minimum-VAL/first-tie checkpoint per arm, no TEST/LIB reselection or per-problem assembly. Actual all18 online native+selector timing is GPU-resident batch128 amortized, NOT singleton or I/O-inclusive end-to-end. Dataset I/O, CPU transfer, training and solver decoding/execution are excluded. Gradient/state witnesses are implementation checks, not selection-effectiveness evidence.

R106/A_frozen | Same selected checkpoint: full TRAIN

Diagnostic only; checkpoint selected on whole18 VAL, SBS fitted on full frozen TRAIN.

Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
ALL	71.8904	66.0332	11.386855	11.400702	11.345073	-0.2202	+0.3645	0.3713
TSP	46.4062	32.8617	8.003923	8.023959	7.970215	-0.2497	+0.4229	0.4176
CVRP	49.0185	18.1284	18.036401	18.151987	17.946816	-0.6368	+0.4992	0.6576
ATSP	79.2457	46.9210	1.638855	1.678679	1.633938	-2.3723	+0.3009	0.2997
OVRP	41.5951	34.2420	7.443505	7.453531	7.397676	-0.1345	+0.6195	0.6321
VRPB	81.6222	80.2765	9.351149	9.357264	9.319342	-0.0653	+0.3413	0.3448
VRPL	38.0358	31.5309	12.392489	12.400882	12.344503	-0.0677	+0.3887	0.3955
VRPTW	80.8333	80.2407	18.809541	18.813703	18.757954	-0.0221	+0.2750	0.2869
OVRPTW	84.1963	83.8025	10.116801	10.118640	10.093220	-0.0182	+0.2336	0.2269
OVRPB	88.7086	88.3580	6.361447	6.363230	6.346658	-0.0280	+0.2330	0.2374
OVRPL	42.3469	34.4716	7.443683	7.453219	7.398483	-0.1280	+0.6109	0.6227
VRPBL	80.7815	79.4593	9.396810	9.403115	9.363196	-0.0670	+0.3590	0.3597
VRPBTW	80.0630	79.4914	19.726236	19.733549	19.642712	-0.0371	+0.4252	0.4098
VRPLTW	80.8951	80.3827	18.766014	18.769959	18.714366	-0.0210	+0.2760	0.2887
OVRPBL	88.8988	88.5556	6.357431	6.358857	6.342513	-0.0224	+0.2352	0.2392
OVRPBTW	83.5679	83.2432	10.622097	10.624446	10.586225	-0.0221	+0.3388	0.3117
OVRPLTW	84.1185	83.7321	10.109410	10.111265	10.085768	-0.0184	+0.2344	0.2292
VRPBLTW	80.0901	79.5630	19.764072	19.770814	19.680364	-0.0341	+0.4253	0.4080
OVRPBLTW	83.6037	83.3370	10.623529	10.625544	10.587365	-0.0190	+0.3416	0.3161

ALL equal-task; ratios of mean costs differ from per-instance actual regret.

R106/A_frozen | Fixed target36 vs R84A

True rescues/losses, false picks and aligned net cost. Historical references are NOT matched.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	2660	65	1867	-0.002305157
VAL	4625	132	2	126	-0.000576724
TEST	4598	122	5	135	-0.001600425

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	0.646502	36566	27140	-0.002305157	-0.023908
VAL	0.250617	1900	1697	-0.000576724	-0.006003
TEST	0.287654	1886	1653	-0.001600425	-0.012981

Raw cost delta is aligned whole-population, equal-task macro; not target-choice frequency.

No matched historical pretraining/compute claim; TEST development-exposed; no rule reselection.

R106/A_frozen | Fixed target36 vs R94B

True rescues/losses, false picks and aligned net cost. Historical references are NOT matched.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	1007	845	1082	0.000564866
VAL	4625	48	33	70	-0.000244210
TEST	4598	39	26	80	0.000009014

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	-0.057476	19946	20784	0.000564866	0.003491
VAL	-0.092593	1080	1155	-0.000244210	-0.001588
TEST	-0.054321	1117	1161	0.000009014	0.002750

Raw cost delta is aligned whole-population, equal-task macro; not target-choice frequency.

No matched historical pretraining/compute claim; TEST development-exposed; no rule reselection.

R106/A_frozen | Fixed target36 vs R102B

True rescues/losses, false picks and aligned net cost. Historical references are NOT matched.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	1167	126	1066	-0.001240228
VAL	4625	53	12	75	-0.000287915
TEST	4598	51	13	67	-0.000922573

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	0.529150	25740	18025	-0.001240228	-0.015410
VAL	0.203704	1286	1121	-0.000287915	-0.003353
TEST	0.297531	1338	1097	-0.000922573	-0.010093

Raw cost delta is aligned whole-population, equal-task macro; not target-choice frequency.

No matched historical pretraining/compute claim; TEST development-exposed; no rule reselection.